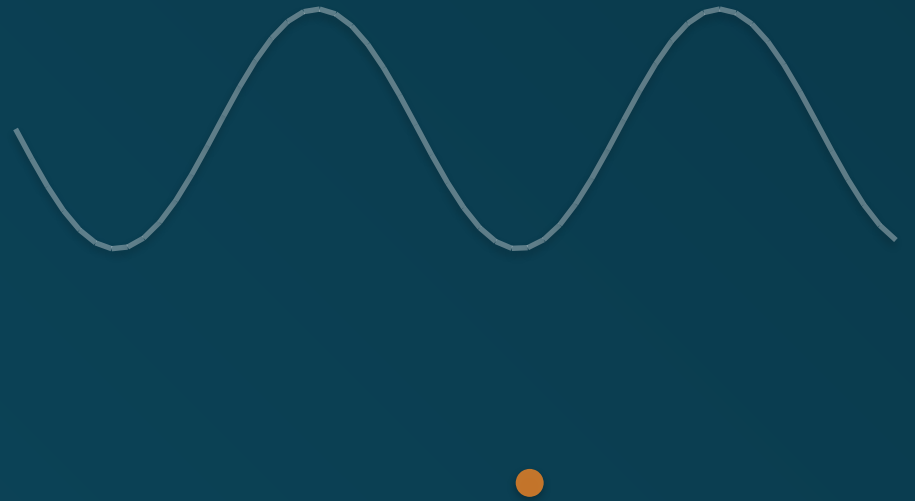


TOPIC 17

Oscillations

振动



CIE 9702 · A Level Physics A2 · Paper 4

剑桥国际物理 A2 分章节讲义 · 中英双语

Learning Outcomes

学习目标

LEARNING OUTCOMES

学习目标

By the end of this topic, you will
be able to...

17.1 Simple harmonic oscillations 简谐振动

- Terms: displacement, amplitude, period, frequency, angular frequency, phase difference 基本术语
- SHM: acceleration $\propto -\text{displacement}$; $a = -\omega^2 x$; $x = x_0 \sin \omega t$ 定义与方程
- Use $v = v_0 \cos \omega t$ and $v = \pm \omega \sqrt{(x_0^2 - x^2)}$; analyse x , v , a graphs 速度与图像

17.2 Energy in s.h.m. 振动能量

- Interchange between kinetic and potential energy 动能势能转化
- Total energy $E = \frac{1}{2}m\omega^2 x_0^2$ 总能量公式

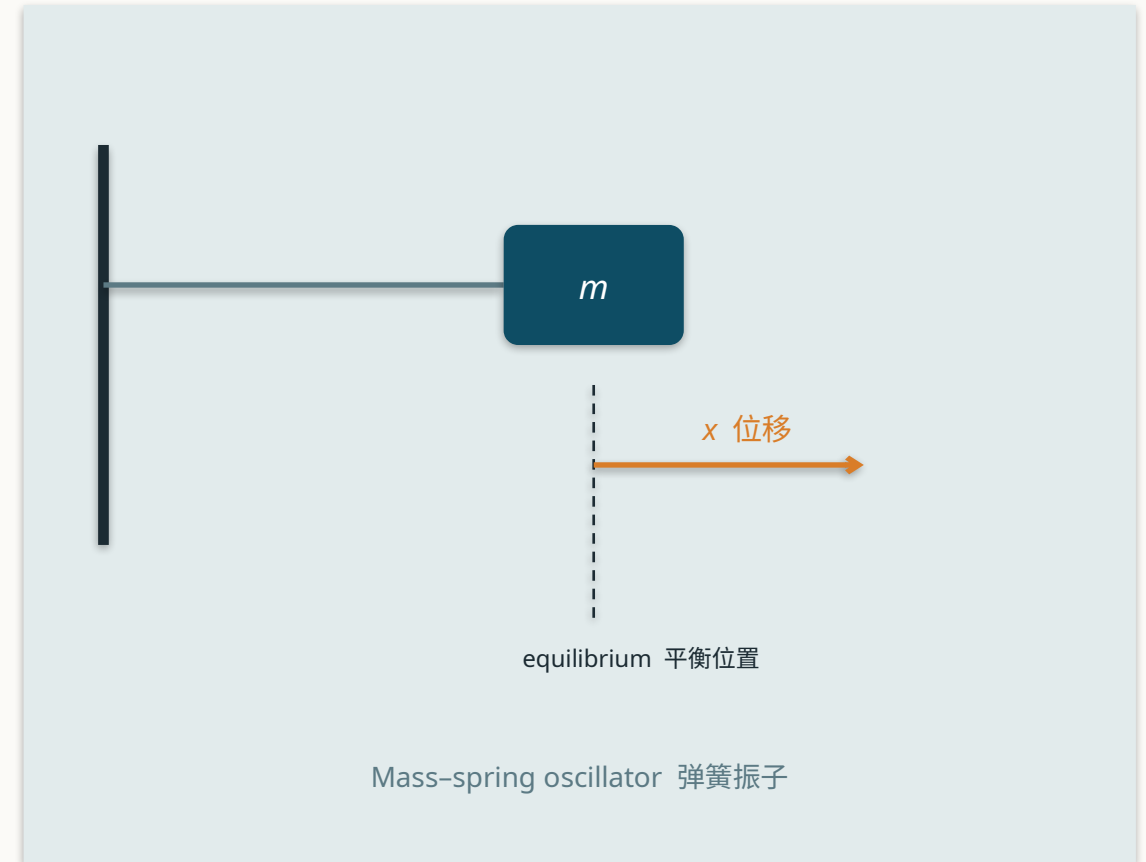
17.3 Damped & forced oscillations, resonance 阻尼与共振

- Light, critical and heavy damping 三种阻尼
- Forced oscillations and resonance 受迫振动与共振

Describing Oscillations

描述振动的物理量

- **Displacement x** 位移 : distance from the equilibrium position
相对平衡位置
- **Amplitude x_0** 振幅 : maximum displacement 最大位移
- **Period T** 周期 / **frequency f** 频率 :
- **Angular frequency** 角频率 : , 单位 rad s^{-1}
- **Phase difference** 相位差 : fraction of a cycle between two oscillations 两振动间的相差



Defining Simple Harmonic Motion

简谐运动的定义

SHM occurs when the acceleration of an object is **proportional to its displacement** from a fixed point, and is always **directed towards that point**.

加速度与位移成正比、方向始终指向平衡位置的振动。

$$a = -\omega^2 x$$

ω^2 为比例常数 · 负号表方向相反

- The minus sign: acceleration always points back to equilibrium 负号 = 回复性
- Restoring force — e.g. spring gives
- Not every oscillation is SHM: check whether $a \propto -x$ 判据: a 是否正比于 $-x$

Equations of SHM

简谐运动方程组

$$x = x_0 \sin \omega t$$

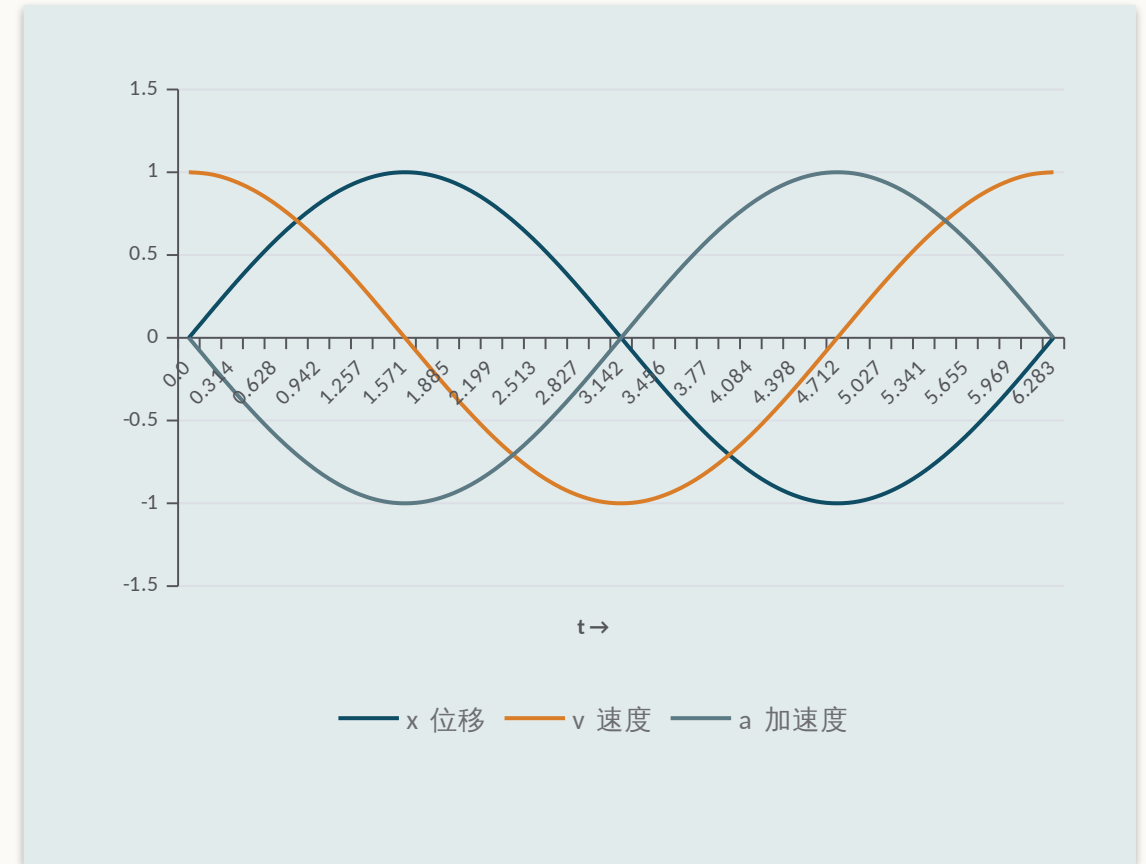
解的形式（从平衡位置计时）

- Velocity 速度：
- Speed at displacement x :
- Acceleration:
- **Maxima:** at $x = 0$; at $x = \pm x_0$
- sin/cos 取决于计时起点: $x = x_0 \cos \omega t$ （从最大位移计时）

Graphs of SHM

x、v、a 图像与相位关系

- v is the gradient of the x-t graph; a is the gradient of the v-t graph 斜率关系
- **v leads x by $T/4$ (90°) ; a is in antiphase with x (180°)**
速度与加速度相位
- At $x = \pm x_0$: $v = 0$, $|a| \text{ max}$; At $x = 0$: $|v| \text{ max}$, $a = 0$ 特殊点对应
- a-x graph: straight line through origin with negative gradient
 $-\omega^2$ a-x 为过原点负斜率直线

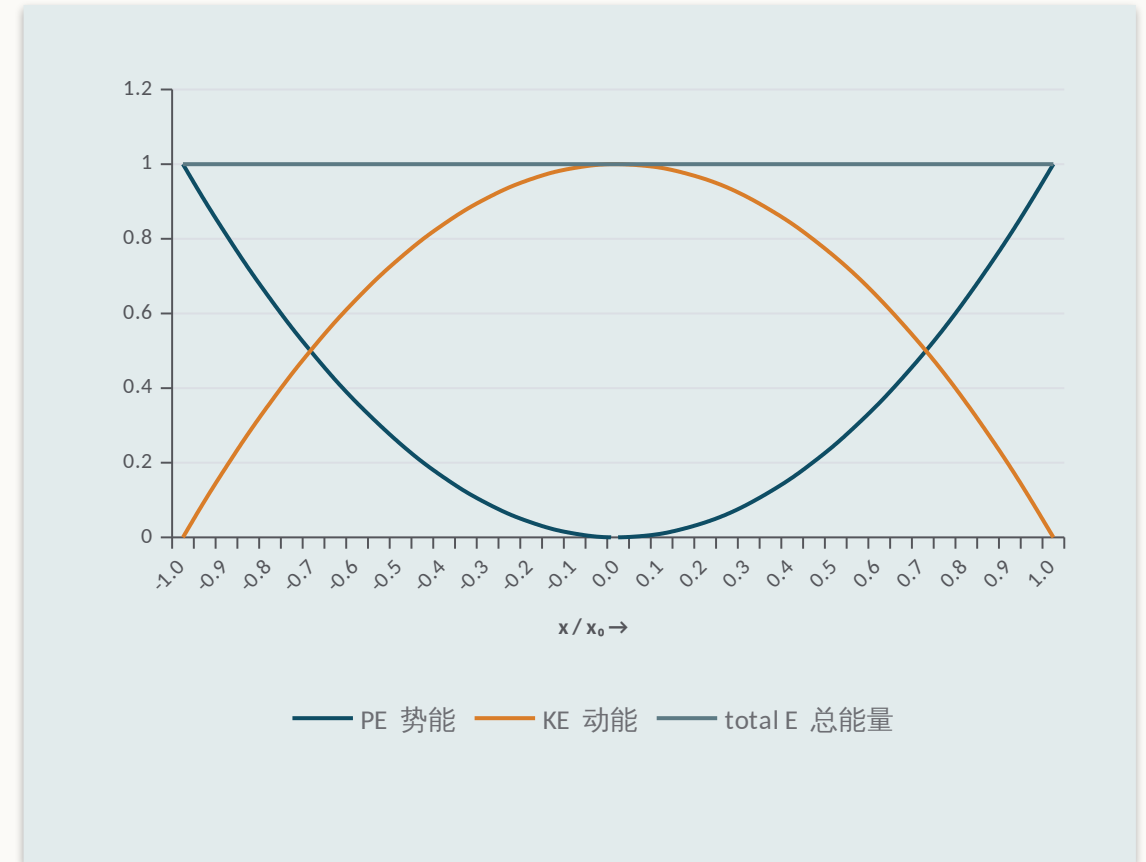


x, v, a against time (normalised) 归一化曲线

Energy in SHM

振动中的能量转化

- Energy interchanges between KE (max at equilibrium) and PE (max at extremes) 动能势能相互转化
- Total energy 总能量：（恒定）
- At displacement x : ,
- $E \propto x_0^2$: doubling the amplitude quadruples the energy 能量与振幅平方成正比



Energy vs displacement 能量 - 位移图 (归一化)

Period of SHM Systems

两种典型振子的周期

- **Mass-spring** 弹簧振子 : restoring force $\Rightarrow$

$$T = 2\pi \sqrt{\frac{m}{k}}$$

mass-spring 弹簧振子周期

- **Simple pendulum** 单摆 (小角度) : restoring force $\Rightarrow$

$$T = 2\pi \sqrt{\frac{l}{g}}$$

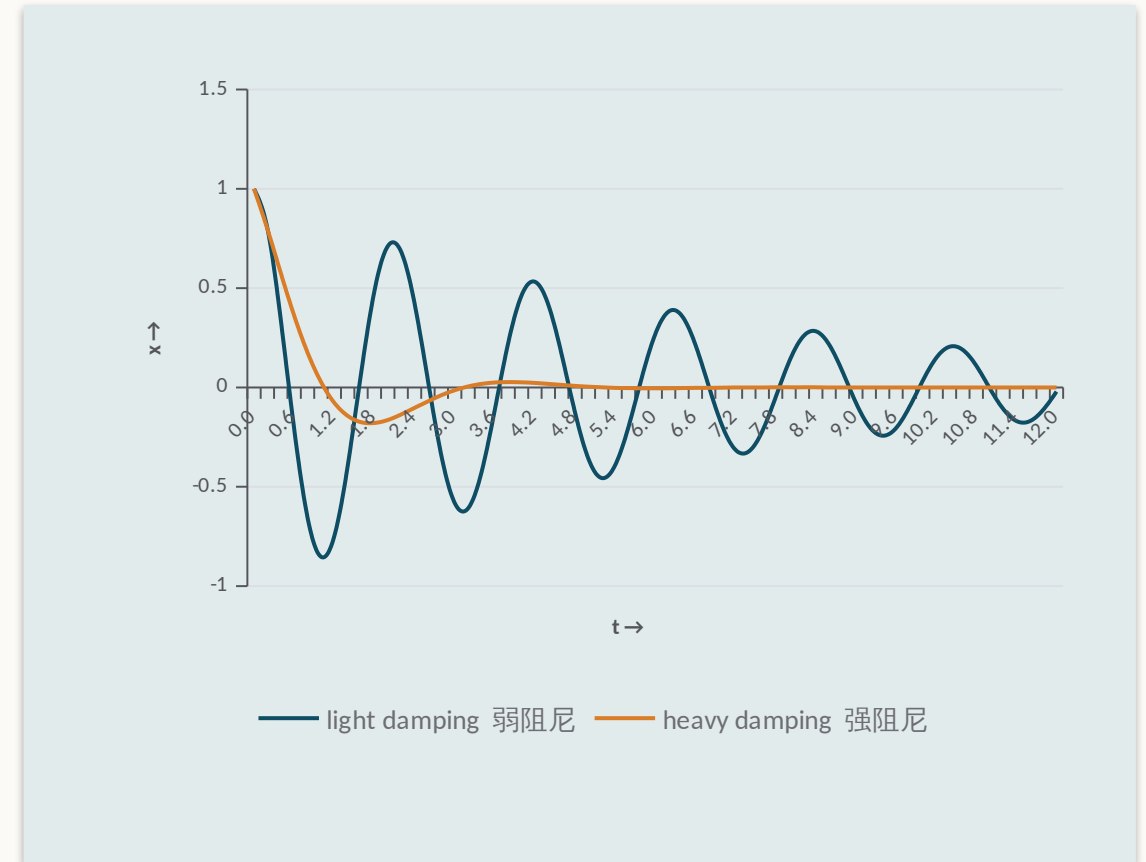
simple pendulum 单摆周期

- Method 通用方法 : write the restoring force, compare with 比较系数得 ω^2
- Period is independent of amplitude (isochronous) 周期与振幅无关

Damping

阻尼振动

- **Damping:** resistive forces dissipate the oscillator's energy as thermal energy 阻力耗散能量
- **Light damping** 弱阻尼 : amplitude decays gradually (exponential envelope) 振幅指数衰减
- **Critical damping** 临界阻尼 : returns to equilibrium in the shortest time without oscillating 最快回平衡无振动
- **Heavy damping** 强 (过) 阻尼 : slow return without oscillating 缓慢蠕回
- Applications 应用 : car shock absorbers (critical), door closers, galvanometers 汽车减震



Displacement-time for damped oscillations 阻尼振动曲线

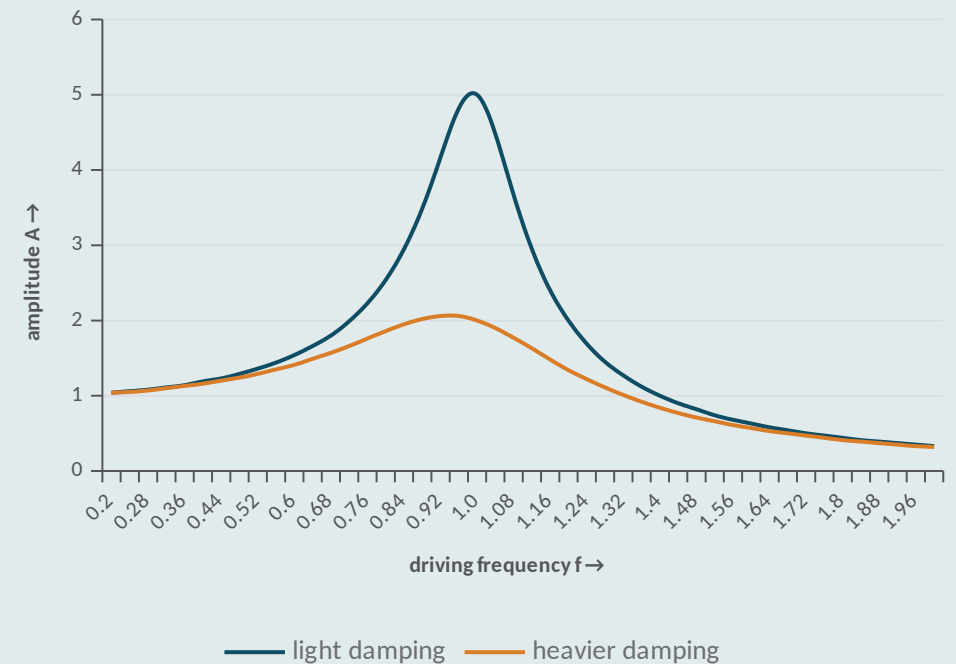
Forced Oscillations & Resonance

受迫振动与共振

Resonance: when the driving frequency equals the natural frequency, the system oscillates with **maximum amplitude**.

驱动频率等于固有频率时振幅最大的现象。

- Forced oscillation: driven by an external periodic force 受迫振动由外界周期力驱动
- Lighter damping $\Rightarrow$ sharper, higher resonance peak 阻尼越小峰越尖越高
- Useful 有益 : radio tuning, microwave ovens, musical instruments
- Harmful 有害 : bridges (Tacoma Narrows), machinery vibration 需避免



Resonance curves 共振曲线

Worked Example 17A

例题解析

WORKED EXAMPLE 17A

The displacement of a particle in SHM is given by $x = 0.36 \sin(1.7t)$ in m, t in s. (a) Find the amplitude, frequency and period. (b) Find the displacement when $t = 0.35$ s.

$\sin$ 内是弧度，计算器必须 RAD 模式！

SOLUTION 解答

Step 1 Compare with $x = A \sin(2\pi f t)$.

Step 2 $A = 0.36$ m, $2\pi f = 1.7$ rad/s.

Step 3 (a) $f = \frac{1.7}{2\pi} = 0.27$ Hz, $T = \frac{1}{f} = 0.59$ s. (b) (calculator in RAD mode 计算器弧度模式)

Answer (a) 0.36 m, 1.7 Hz, 0.59 s (b) 0.12 m

Worked Example 17B

例题解析

WORKED EXAMPLE 17B

A particle of mass 95 g oscillates in SHM with angular frequency 12.5 rad s^{-1} and amplitude 16 mm.

Calculate (a) the total energy; (b) the kinetic and potential energies at half-amplitude ($x = 8.0 \text{ mm}$).

半振幅处势能占 1/4、动能占 3/4 (因 $E \propto x^2$)。

SOLUTION 解答

Step 1 Convert: $m = 0.095 \text{ kg}$, $x = 0.008 \text{ m}$.

Step 2 (a)

Step 3 (b) ; $x = x_0/2 \Rightarrow$

Step 4

Answer (a) (b) ,

Class Practice

随堂练习

- Q1** A particle oscillates in SHM with frequency 4.0 Hz and amplitude 8.0 cm; its mass is 0.35 kg. For displacement $x = 7.0$ cm, calculate (a) the kinetic energy; (b) the potential energy; (c) the total energy.
- Q2** State the defining condition for SHM, and explain why a bouncing ball is NOT in simple harmonic motion. SHM 判据辨析。
- Q3** Sketch (or describe) how the amplitude of a forced oscillator varies with driving frequency for light and for heavy damping, and mark the resonance frequency. 共振曲线描述。

Answers Q1 (a) 0.17 J (b) 0.54 J (c) 0.71 J Q2 a 必须正比于 $-x$; 弹球受恒力 (重力) 回复力不随位移变化 Q3 轻阻尼峰更高更尖, 峰值在固有频率处

Summary

本章小结

KEY FORMULAS 核心公式

$$a = -\omega^2 x$$

SHM definition 定义

$$x = x_0 \sin \omega t$$

displacement 位移

$$v = \pm \omega \sqrt{x_0^2 - x^2}$$

velocity 速度

$$E = \frac{1}{2} m \omega^2 x_0^2$$

total energy 总能量

$$T = 2\pi \sqrt{m/k}$$

mass-spring 弹簧

$$T = 2\pi \sqrt{l/g}$$

pendulum 单摆

KEY IDEAS 核心思想

- $a \propto -x$ and directed to equilibrium — the defining test of SHM 判据：加速度 $\propto$ -位移。
- v leads x by 90° ; a is in antiphase with x 相位关系常考。
- Damping reduces amplitude; resonance occurs at the natural frequency 阻尼减幅，共振在固有频率。

Exam Tips

备考提示

- Angles are in **radians**: set your calculator to RAD mode for $\sin \omega t$. 计算器开弧度。
- Read graphs via gradients: $v = \text{gradient of } x-t$; $a = \text{gradient of } v-t$. Maximum speeds at $x = 0$. 图像斜率含义。
- To prove SHM, derive the restoring force and show $\ddot{x} = -\omega^2 x$, then identify ω . 证明 SHM 要写恢复力表达式。
- Light damping barely changes the period but reduces amplitude exponentially. 轻阻尼对周期影响小。